RealityCraft

Can AI Know What It Knows?

A Beginner's Guide to Investigating Philosophical Questions with AI Agents

Project Overview

In this project, you will work with a group to investigate a philosophical question about artificial intelligence (AI). Rather than simply discussing what you think the answer is, you will design a small experiment to explore the question.

You might investigate whether an AI can know something it has not seen for itself, whether it can be fooled by misleading information, or whether an AI saying something is wrong is the same as a human believing it is wrong.

You do not need to be an expert in philosophy, computer science, or statistics. The goal is to practice asking good questions, developing arguments, examining evidence, and thinking carefully about what your conclusions actually show.

You will use AI agents in a shared virtual environment, such as Minecraft, to test your ideas. Your project should connect a philosophical debate to an experiment that can provide evidence relevant to that debate.

1. Choose a Philosophical Question

Start with a question that genuinely interests you. A good philosophical question is one that people can reasonably disagree about and that requires more than a simple factual answer.

Here are some possibilities:

- **Knowledge:** Can you know something without seeing it for yourself?
- **Trust:** Should you trust information just because someone tells you it is true?
- **Belief:** Is saying that something is true the same as believing it is true?
- **Deception:** Can an AI be fooled into believing something false?

- **Reality:** How can you know that the world you experience is real?
- **Morality:** Is there a meaningful difference between a human and an AI saying that stealing is wrong?
- **Understanding:** Does an AI really understand what it says, or is it simply producing the right words?

Choose one main question. You do not need to solve an entire philosophical debate. Instead, identify one small part of the question that you can investigate.

Your research question: What exactly do you want to find out?

Your initial answer: What do you currently think the answer might be, and why?

2. Explain the Philosophical Debate

Before designing your experiment, explain the philosophical problem in your own words.

Address these questions:

- What does your main question mean?
- Why is it difficult to answer?
- What are two different ways someone might answer it?
- What reasons might someone give for each answer?
- What would count as evidence for or against each position?

For example, consider the question, "Can you know something without seeing it for yourself?"

One person might argue that you can know something through reliable testimony. If a trusted friend tells you where a missing object is, you may have good reason to believe them.

Another person might argue that hearing something from another person is not enough. Your friend could be mistaken, dishonest, or misinformed.

Neither position should be dismissed without considering its reasons.

You should use at least one relevant philosophical reading or course concept to help explain the debate. Your goal is not simply to state your opinion, but to show that you understand the problem and the arguments surrounding it.

3. Turn Your Question into a Testable Hypothesis

An experiment cannot directly answer every philosophical question. However, it can help us examine specific claims and determine whether certain evidence supports them.

A **hypothesis** is a prediction about what you expect to happen.

Suppose your question is whether an AI agent can gain knowledge from another agent's testimony.

You might predict:

"If an AI agent receives accurate information from another agent, it will be able to use that information to find the correct object, even when it cannot see the object for itself."

Now identify what your experiment would need to measure.

Question	Example answer
What are we investigating?	Whether an AI can use information provided by another agent
What do we predict will happen?	The agent will find the correct object after receiving accurate information
What will we observe?	Whether the agent goes to the correct location
What would count as success?	The agent correctly identifies or retrieves the object
What else might explain success?	The agent may have guessed or received the answer from another source

Be specific. If you are investigating knowledge, explain why the behavior you measure might be relevant to knowledge—and why it might not be enough to establish knowledge by itself.

4. Design Your Experiment

Your experiment should allow you to compare what happens under different conditions.

For example, imagine that one AI agent needs to find a chest in Minecraft.

You could test three situations:

Condition A: Information from another agent

The first agent tells the second agent where the chest is. The second agent cannot directly see the chest.

Condition B: Direct access

The second agent can see the chest and must find it.

Condition C: No information

The second agent receives no information about the chest's location and must try to find it.

These conditions help you compare the agent's ability to use information with its ability to find the chest independently.

For your own experiment, explain:

- What will the AI agents be asked to do?
- What information will each agent receive?
- What will you change between conditions?
- What will remain the same?
- How will you determine whether the agent succeeds?
- How will you prevent the agent from receiving clues that give away the answer?

Try to change only one important factor at a time. Otherwise, it may be difficult to tell what caused the results.

If you cannot run a computer experiment, ask your instructor whether you may use a structured thought experiment or analyze an existing AI demonstration instead.

5. Collect and Describe Your Evidence

After running your experiment, record what happened. Do not record only the results that support your original prediction.

Include:

- How many trials you conducted.
- How often the agent succeeded or failed.
- Any important differences between the conditions.
- Examples of the agent's messages or actions.
- Unexpected results or problems during testing.

A simple table might look like this:

Condition	Successful attempts	Total attempts	Success rate
Information from another agent	_____	_____	_____%
Direct access	_____	_____	_____%
No information	_____	_____	_____%

Calculate the success rate by dividing the number of successful attempts by the total number of attempts and multiplying by 100.

If you did not run the experiment, clearly identify your evidence as coming from a thought experiment, demonstration, or previously published study.

Important: Never invent results to make your argument more convincing. If your experiment does not work as expected, that is still something worth discussing.

6. Explain What Your Results Mean

This is the most important part of the project. Your job is not just to describe what the AI did. You must explain how the results relate to your philosophical question.

Consider the following questions:

- Did the results match your prediction?
- What do the results suggest about your original question?
- Is there another explanation for what happened?
- What can your experiment establish, and what can it not establish?
- Would someone who disagrees with your position interpret the results differently?

For example, if an AI agent successfully finds a chest after another agent tells it where to look, that shows it can use the information to complete the task.

But does it show that the agent *knows* where the chest is?

Perhaps. However, the agent might simply be following instructions, repeating information, or acting on a pattern without possessing knowledge in the sense philosophers usually discuss.

You must explain why you think the evidence supports your conclusion and acknowledge reasonable objections.

Remember: An experiment can support a philosophical argument without settling the entire philosophical debate.

7. Consider an Objection

A strong philosophical investigation takes opposing arguments seriously.

Imagine that someone disagrees with your conclusion. What is the strongest objection they could make?

For example:

"Your experiment shows only that the AI can follow directions. It does not show that the AI understands the information or knows that it is true."

Respond to the objection:

1. Explain why the objection is reasonable.
2. Describe how you would answer it.
3. Identify what additional evidence might help resolve the disagreement.

You do not have to defeat every objection. Recognizing the limits of your own argument is part of doing philosophy well.

8. Write Your Conclusion

End by answering your original question as clearly as you can.

Your conclusion should include:

- Your main finding.
- The evidence that supports your interpretation.
- How the results relate to the philosophical debate.
- One important limitation of your investigation.
- One new question you would investigate next.

Avoid claiming that your experiment proves more than it actually does.

For example, rather than writing, "We proved that AI can think," you might write, "Our results show that the AI agent can use information provided by another agent to complete a task. Whether this ability counts as genuine understanding remains an open philosophical question."

A thoughtful conclusion can identify an important uncertainty rather than provide a final answer.

9. Suggested Paper Structure

Organize your final paper using the following sections.

Title: Create a short, memorable title that communicates your philosophical question and experiment.

1. Introduction - Introduce your philosophical question. - Explain why it matters. - State your initial position or hypothesis. - Briefly describe your experiment.

2. Philosophical Background - Explain the main concepts. - Present at least two perspectives on the question. - Discuss a relevant course reading or philosophical argument.

3. Experiment Design - Describe the AI agents and virtual environment. - Explain the different conditions. - Identify what you measured and how you measured it.

4. Results - Report what happened. - Include a simple table or figure if appropriate. - Describe important examples and unexpected findings.

5. Philosophical Discussion - Explain what the results suggest. - Consider alternative explanations. - Present an objection and your response. - Discuss what the evidence cannot establish.

6. Conclusion - Answer your original question as far as the evidence allows. - Identify a limitation. - Suggest a question for further investigation.

References - List the philosophical readings and other sources you used. - Follow your instructor's preferred citation style.

10. What Makes a Strong Project?

A successful project does not need a complicated experiment or a surprising discovery. It needs a clear question, careful reasoning, honest use of evidence, and a thoughtful conclusion.

Before submitting, ask yourself:

- ☐ Is my philosophical question clear and specific?
- ☐ Have I explained the issue in my own words?
- ☐ Have I considered more than one position?
- ☐ Does my experiment test something relevant to the question?
- ☐ Have I explained how I collected and interpreted my evidence?
- ☐ Have I considered alternative explanations?
- ☐ Have I included a serious objection to my conclusion?
- ☐ Have I acknowledged the limits of what I can claim?
- ☐ Have I cited my sources and represented them accurately?

Final Reminder

Philosophy is not just about having opinions. It is about asking meaningful questions, giving reasons for your answers, examining evidence, and being willing to reconsider your position.

AI gives us a new way to explore some very old philosophical problems. Your task is to use that opportunity carefully: investigate a question, make an argument, test your assumptions, and explain what you learned.